

# Code-Layout, Readability and Reusability

|               |                                                         |
|---------------|---------------------------------------------------------|
| Date          | 30 September 2026                                       |
| Team ID       | [ENTER TEAM ID]                                         |
| Project Name  | ComicCraft - AI Comic Story Creator using Gemini Models |
| Maximum Marks | 5 Marks                                                 |

This document evaluates the structure, readability and reusability of the ComicCraft codebase.

## Code Layout Checklist

| S.No | Code Quality Parameter    | Description                                  | Followed (Yes / No / Partial) | Remarks                                                                                                                       |
|------|---------------------------|----------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 1    | Consistent Indentation    | Uniform spacing used throughout              | Yes                           | 4-space PEP 8 indentation in all Python files; 2-space in HTML templates.                                                     |
| 2    | Proper File Structure     | Files and folders logically organised        | Yes                           | app/ (routes, config, schemas), app/services/ (one file per AI or export step), templates/, static/, tests/, scripts/.        |
| 3    | Meaningful Variable Names | Variables reflect their purpose              | Yes                           | e.g. story_prompt, character_name, image_paths, pdf_path, layout.                                                             |
| 4    | Function / Method Names   | Functions descriptively named                | Yes                           | generate_outline, generate_story, generate_image, build_comic_layout, save_pdf, create_comic (names match the project brief). |
| 5    | Code Comments             | Inline and block comments explain logic      | Yes                           | Module and function docstrings plus short inline comments for non-obvious logic (retry, JSON cleaning, locks).                |
| 6    | Modular Design            | Code split into reusable functions / modules | Yes                           | Each workflow step is its own module; pipeline.create_comic() composes them.                                                  |
| 7    | No Redundant Code         | Duplicate or unused code removed             | Yes                           | Form route and JSON route share create_comic() and PromptRequest; one shared Gemini client.                                   |
| 8    | Error Handling            | Exceptions handled gracefully                | Yes                           | GenerationError, Gemini retries, validation errors (422), 502/500 pages, placeholder image fallback.                          |

## Reusable Components / Modules

| S.No | Component / Module Name                   | Language / Technology     | Where Reused                                                  | Reusability Level (High / Medium / Low) |
|------|-------------------------------------------|---------------------------|---------------------------------------------------------------|-----------------------------------------|
| 1    | services/gemini_client.call_gemini()      | Python, google-genai      | gemini_flash and gemini_pro (any other Gemini task)           | High                                    |
| 2    | services/pipeline.create_comic()          | Python                    | /generate form route and /generate-comic/json API route       | High                                    |
| 3    | schemas.PromptRequest                     | Python, Pydantic          | Form validation and JSON API validation; auto-generated /docs | High                                    |
| 4    | services/image_generator.generate_image() | Python, Diffusers, Pillow | Pipeline (each panel) and /test-image route                   | High                                    |
| 5    | services/exporters.save_pdf()             | Python, FPDF2             | Comic export; can export any layout list                      | Medium                                  |

| S.No | Component / Module Name              | Language / Technology | Where Reused                                        | Reusability Level (High / Medium / Low) |
|------|--------------------------------------|-----------------------|-----------------------------------------------------|-----------------------------------------|
| 6    | templates (Jinja2) with shared style | HTML, CSS, Jinja2     | index, preview and success pages share visual style | Medium                                  |

### Overall Code Quality Assessment

| Aspect                   | Rating (1-5) | Comments                                                                           |
|--------------------------|--------------|------------------------------------------------------------------------------------|
| Code Layout & Structure  | 5            | Clear separation of routing, services, templates and tests.                        |
| Readability              | 4            | Small functions with docstrings; some prompt strings are long.                     |
| Reusability              | 4            | Shared client, pipeline and schema; templates duplicate some CSS.                  |
| Documentation / Comments | 4            | README, docstrings and .env.example; no auto-generated API reference beyond /docs. |
| Overall Score            | 4.25         | Average of the four ratings above (self-assessment).                               |